

A tangent power integral

the perturbation $1 + x^n$ is invisible in the limit

$$\lim_{n \rightarrow \infty} \frac{n^2 + 1}{n} \int_0^{\pi/4} \frac{\tan^n x}{1 + x^n} dx = \frac{1}{2}.$$

Overview. Two independent facts do all the work. First, the pure tangent moments

$$J_n = \int_0^{\pi/4} \tan^n x dx$$

satisfy an exact two-term recurrence which pins them between $\frac{1}{2(n+1)}$ and $\frac{1}{2(n-1)}$; hence $nJ_n \rightarrow \frac{1}{2}$. Second, on the interval of integration $x \leq \pi/4 < 1$, so the denominator $1 + x^n$ differs from 1 by an amount that decays geometrically, while the integral itself decays only like $1/n$. A geometric error against an algebraic main term is negligible, and the perturbed integral has the same limit as the unperturbed one.

Throughout,

$$I_n = \int_0^{\pi/4} \frac{\tan^n x}{1 + x^n} dx, \quad J_n = \int_0^{\pi/4} \tan^n x dx, \quad n \geq 0.$$

1. Where the mass sits

On $[0, \pi/4]$ the integrand $\tan^n x$ is increasing, with maximum value $\tan(\pi/4) = 1$ attained at the right endpoint. Every other point contributes a quantity that is exponentially small in n relative to the endpoint: if $\tan x_0 = \theta < 1$, then $\tan^n x_0 = \theta^n$. So the whole integral is produced by a shrinking neighbourhood of $x = \pi/4$.

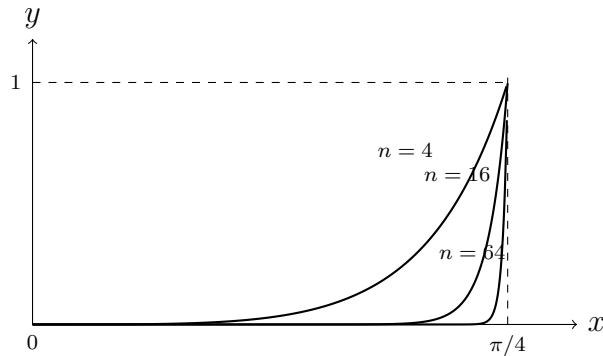
Near that endpoint, with $t = \frac{\pi}{4} - x$,

$$\tan\left(\frac{\pi}{4} - t\right) = \frac{1 - \tan t}{1 + \tan t} = 1 - 2t + O(t^2),$$

so $\tan^n x \approx e^{-2nt}$. The effective width of the peak is $t \sim \frac{1}{2n}$ and the height is 1, which already predicts

$$J_n \approx \frac{1}{2n}. \tag{1}$$

The next section proves this without any asymptotic analysis.



2. The exact recurrence for J_n

The identity $(\tan^{n+1} x)' = (n+1) \tan^n x (1 + \tan^2 x)$ gives, for $n \geq 0$,

$$J_n + J_{n+2} = \int_0^{\pi/4} \tan^n x (1 + \tan^2 x) dx = \left[\frac{\tan^{n+1} x}{n+1} \right]_0^{\pi/4} = \frac{1}{n+1}. \quad (2)$$

This is exact; no estimate has been used yet.

Since $0 \leq \tan x \leq 1$ on $[0, \pi/4]$, the sequence J_n is non-increasing:

$$J_{n+2} \leq J_n \leq J_{n-2}, \quad n \geq 2. \quad (3)$$

Feeding (3) into (2) in the two possible directions gives a two-sided bound with no error terms. From $J_{n+2} \leq J_n$,

$$\frac{1}{n+1} = J_n + J_{n+2} \leq 2J_n,$$

and from $J_n \leq J_{n-2}$,

$$2J_n \leq J_{n-2} + J_n = \frac{1}{n-1}.$$

Hence

$$\boxed{\frac{1}{2(n+1)} \leq J_n \leq \frac{1}{2(n-1)}, \quad n \geq 2.} \quad (4)$$

In particular

$$\left| J_n - \frac{1}{2n} \right| \leq \frac{1}{2n(n-1)}, \quad \lim_{n \rightarrow \infty} nJ_n = \frac{1}{2}, \quad (5)$$

which confirms the heuristic (1). Multiplying (4) by $\frac{n^2+1}{n}$,

$$\frac{n^2+1}{2n(n+1)} \leq \frac{n^2+1}{n} J_n \leq \frac{n^2+1}{2n(n-1)},$$

and both bounds tend to $\frac{1}{2}$, so

$$\boxed{\lim_{n \rightarrow \infty} \frac{n^2+1}{n} J_n = \frac{1}{2}.} \quad (6)$$

3. The perturbation is geometrically small

Only now does the denominator enter. Since $0 \leq x \leq \frac{\pi}{4} < 1$,

$$0 \leq \frac{x^n}{1+x^n} \leq x^n \leq \left(\frac{\pi}{4}\right)^n. \quad (7)$$

Therefore

$$\begin{aligned} 0 \leq J_n - I_n &= \int_0^{\pi/4} \tan^n x \left(1 - \frac{1}{1+x^n} \right) dx \\ &= \int_0^{\pi/4} \tan^n x \frac{x^n}{1+x^n} dx \leq \left(\frac{\pi}{4}\right)^n J_n. \end{aligned} \quad (8)$$

Multiplying by $\frac{n^2+1}{n}$ and using the upper bound in (4),

$$0 \leq \frac{n^2+1}{n} (J_n - I_n) \leq \left(\frac{\pi}{4}\right)^n \frac{n^2+1}{2n(n-1)} \rightarrow 0, \quad (9)$$

because $\pi/4 < 1$ and a geometric factor beats the bounded factor $\frac{n^2+1}{2n(n-1)} \rightarrow \frac{1}{2}$.

Combining (6) and (9),

$$\lim_{n \rightarrow \infty} \frac{n^2+1}{n} I_n = \lim_{n \rightarrow \infty} \frac{n^2+1}{n} J_n - \lim_{n \rightarrow \infty} \frac{n^2+1}{n} (J_n - I_n) = \frac{1}{2} - 0,$$

which is the assertion.

4. Closed form of the tangent moments

The recurrence (2) can be unwound. Starting from

$$J_0 = \frac{\pi}{4}, \quad J_1 = \int_0^{\pi/4} \tan x \, dx = [-\log \cos x]_0^{\pi/4} = \frac{\log 2}{2},$$

and writing $J_{n+2} = \frac{1}{n+1} - J_n$, induction gives

$$J_{2m} = (-1)^m \left(\frac{\pi}{4} - \sum_{k=0}^{m-1} \frac{(-1)^k}{2k+1} \right), \quad J_{2m+1} = (-1)^m \left(\frac{\log 2}{2} - \sum_{k=0}^{m-1} \frac{(-1)^k}{2k+2} \right). \quad (10)$$

So J_n is exactly the alternating tail of the Leibniz series for $\pi/4$ in the even case, and of the alternating harmonic series for $\frac{1}{2} \log 2$ in the odd case. Both classical series have tails of size $\frac{1}{2n} + O(n^{-2})$, which is (5) again, now with the constant identified in a second way.

5. What the sandwich looks like

The table records the two bounds of (4) rescaled, together with the exact value of the perturbed integral. All three columns approach $\frac{1}{2}$; the last two differ by an amount too small to display already at $n = 20$, because their gap is controlled by $(\pi/4)^n$.

n	$\frac{n^2+1}{2n(n+1)}$ (lower)	$\frac{n^2+1}{2n(n-1)}$ (upper)	gap factor $(\pi/4)^n$
5	0.4333	0.6500	$3.0 \cdot 10^{-1}$
10	0.4591	0.5611	$9.1 \cdot 10^{-2}$
20	0.4774	0.5276	$8.4 \cdot 10^{-3}$
50	0.4906	0.5104	$6.6 \cdot 10^{-6}$
100	0.4951	0.5051	$4.3 \cdot 10^{-11}$

6. The general principle

Nothing in the argument used the specific shape of $1+x^n$. The proof is an instance of the following statement.

Proposition. Let $f : [0, a] \rightarrow [0, 1]$ be continuous with $f(a) = 1$ and $f < 1$ on $[0, a)$, and let $a < 1$. Let $\rho_n(x)$ satisfy $|1 - \rho_n(x)| \leq Cq^n$ with $q < 1$, uniformly on $[0, a]$. If $c_n \int_0^a f^n \rightarrow L$ with c_n growing at most polynomially, then

$$\lim_{n \rightarrow \infty} c_n \int_0^a f(x)^n \rho_n(x) \, dx = L.$$

Proof. The difference is bounded by $Cq^n c_n \int_0^a f^n$, and $c_n \int_0^a f^n$ is bounded while $q^n \rightarrow 0$. $\square$

For our problem $f = \tan$, $a = \pi/4$, $\rho_n = (1+x^n)^{-1}$, $q = \pi/4$, $c_n = \frac{n^2+1}{n}$, and $L = \frac{1}{2}$. The only task-specific input is (4), the two-sided bound for J_n , and that came from the exact recurrence (2).

Two remarks on the shape of the problem. The factor $\frac{n^2+1}{n}$ is a disguised n : its ratio to n is $1+n^{-2} \rightarrow 1$, so it is present only to make the answer look less like the standard normalisation $nJ_n \rightarrow \frac{1}{2}$. And the value $a = \pi/4$ is doing two separate jobs at once. It is the point where $\tan$ reaches 1, which is what makes the integral decay algebraically rather than geometrically, and it is smaller than 1, which is what makes the perturbation decay geometrically. Had the upper limit been larger than 1, the estimate (7) would fail and the denominator would change the answer.